BIO 304 . HUMAN ANATOMY & PHYSIOLOGY . WEEK 17 . MODULE 14 OF 17

Module 14: Respiratory System

Topics: Respiratory Anatomy and Mechanics, Gas Exchange and Transport

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TOPIC 1 OF 2 . WEEK 17 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

Respiratory Anatomy and Mechanics

Conducting vs respiratory zones, and how pressure changes move air in and out.

The Science

Two zones. Conducting zone: nose, pharynx, larynx, trachea, bronchi, bronchioles. Warms, humidifies, filters. No gas exchange happens here. Respiratory zone: respiratory bronchioles, alveolar ducts, alveoli. Gas exchange happens here. The alveoli are simple squamous (Type I pneumocytes) with surfactant from Type II pneumocytes.

Pleura: parietal pleura lines the thoracic cavity; visceral pleura covers the lung. Between them is a tiny pleural cavity with serous fluid and (critically) negative pressure. The negative pressure couples the lung to the chest wall: as the chest wall expands during inspiration, the lung is pulled outward with it. Lose the negative pressure (pneumothorax) and the lung collapses elastically.

Mechanics: Boyle's law. Pressure and volume vary inversely at constant temperature. Inspiration: diaphragm contracts (pulls down) and external intercostals lift the rib cage; thoracic volume rises; alveolar pressure drops below atmospheric; air flows in. Expiration at rest is passive: muscles relax, elastic recoil shrinks the thorax, alveolar pressure rises above atmospheric, air flows out. Forced expiration adds internal intercostals and abdominal muscles.

Teaching This Topic

Before the video (drawing prompt)

Have students sketch the airway from mouth to alveolus, label conducting vs respiratory zone, and identify Type I and Type II pneumocytes at the alveolus.

Common misconceptions to address

- Students think the lungs actively pull air in. Air is sucked in passively by the pressure gradient produced by the diaphragm and chest wall.
- Surfactant is thought of as optional. It is essential: without it, alveolar surface tension collapses the alveoli.
- Pleural pressure is sometimes assumed to equal atmospheric. It is normally slightly negative; that is what holds the lung expanded.

Order of operations (what to teach first)

Anatomy and the two zones, then the pleura and the negative pressure rule, then Boyle's law and the mechanics of breathing.

Self-test prompts

- What is the main muscle of inspiration?
- What does surfactant do?
- Why does a pneumothorax cause the lung to collapse?

Why This Matters to Your Patients

Pneumothorax is air in the pleural cavity (from trauma, spontaneous rupture, or iatrogenic); the affected lung collapses; treatment is chest tube drainage to restore negative pressure. Pleural effusion is fluid in the pleural cavity (transudate from heart failure or hypoalbuminemia, exudate from infection or malignancy). Asthma is bronchospasm and airway inflammation, narrowing the conducting zone. COPD (emphysema and chronic bronchitis) destroys alveoli and obstructs airways. Neonatal respiratory distress syndrome is surfactant deficiency in premature infants; exogenous surfactant therapy is life-saving. Every breath in your patient is a Boyle's-law transaction.

TOPIC 2 OF 2 . WEEK 17 . 12 CARDS (4 DOK 1 / 4 DOK 2 / 4 DOK 3)

Gas Exchange and Transport

Diffusion at the alveoli and tissues, and how O■ and CO■ ride in the blood.

The Science

Gas exchange happens by diffusion down partial pressure gradients. External respiration: between alveolar air and pulmonary capillary blood. Alveolar O_2 is about 100 mmHg; pulmonary venous blood arrives at 40 mmHg; O_2 diffuses into blood until equilibrium. CO_2 goes the other way. Internal respiration: between systemic capillary blood and tissue. The gradients are reversed there.

Oxygen transport: about 98% bound to hemoglobin, 2% dissolved in plasma. Each Hb molecule binds 4 O_2 cooperatively (the binding of one O_2 increases affinity for the next). The Hb-O_2 dissociation curve is sigmoidal. Conditions that shift it right (acidosis, high CO_2, high temperature, high 2,3-BPG: the Bohr effect) cause Hb to release MORE O_2 to tissues. This is why exercising muscle, which is acidic, hot, and high in CO_2, gets extra oxygen automatically.

CO_2 transport: about 70% as bicarbonate (the carbonic anhydrase reaction in RBCs: $CO_2 + H_2O \rightarrow H_2CO_3 \rightarrow H^+ + HCO_3^-$), 20% bound to hemoglobin, 10% dissolved. The bicarbonate system is also the body's main blood buffer; respiratory rate adjusts CO_2 and therefore pH on a minute-to-minute basis.

Control of ventilation: central chemoreceptors in the medulla respond to CSF pH (a proxy for arterial CO_2). Peripheral chemoreceptors (carotid and aortic bodies) respond to severely low arterial O_2 (typically below 60 mmHg). CO_2 is the primary moment-to-moment driver. In chronic CO_2 retainers (severe COPD), the central drive desensitizes and the patient relies on the hypoxic drive; high inspired O_2 can blunt this and cause hypoventilation.

Teaching This Topic

Before the video (drawing prompt)

Have students predict: at exercising muscle, does Hb release more or less O_2? Why? They have to commit before the Bohr effect is explained.

Common misconceptions to address

- Students think dissolved O_2 is the main carrier. No, 98% is on Hb. Dissolved O_2 (measured as pO_2) is just the part that determines Hb saturation.
- CO_2 is thought to be transported mainly bound to Hb. No, mostly as bicarbonate.
- Students think low O_2 is the main driver of breathing. It is not. CO_2 (via pH of the CSF) is the primary minute-to-minute driver.

Order of operations (what to teach first)

Where exchange happens (external vs internal), then O_2 transport with the Bohr effect, then CO_2 transport with the bicarbonate buffer, then control of ventilation.

Self-test prompts

- What is the main form of CO_2 transport in blood?
- What conditions shift the Hb-O_2 curve to the right and what is the consequence?
- Why does CO poisoning cause hypoxia even though arterial pO_2 is normal?

Why This Matters to Your Patients

Carbon monoxide poisoning: CO binds Hb with 250x O_2 affinity; oxygen-carrying capacity collapses, even though pO_2 looks normal on blood gas. Pulse oximetry can be falsely normal; co-oximetry is required. Treatment is 100% O_2 and, in severe cases, hyperbaric oxygen. Acute respiratory acidosis (CO_2 retention from respiratory failure) produces falling pH; renal compensation takes days. Chronic respiratory acidosis (COPD) has compensated pH because the kidneys have retained HCO_3^- over time. Pulse oximetry, arterial blood gas interpretation, and supplemental O_2 titration are core nursing skills that depend on this physiology.